HPC: Access to Baobab cluster

Parallelism TP

October 14, 2026

Connection

Access to the cluster:

- From the browser: https://openondemand.baobab.hpc.unige.ch/. Then "Cluster" - "Shell Access"
- From the terminal: `ssh <user>@login1.baobab.hpc.unige.ch`

File copy

Copy files from the terminal by using `scp`:

- From the local to the distant machine:

  `scp <path1> <user>@login1.baobab.hpc.unige.ch:<path2>`
- From the distant to the local machine:

  `scp <user>@login1.baobab.hpc.unige.ch:<path1> <path2>`

Example:

```
scp ./my_folder/foo.txt \
rambossm@login1.baobab.hpc.unige.ch:~/
```

From the browser: Files - Home Directory - Upload

Job: Modules

Many programs are pre-installed by the HPC team: access using the modules.

- Module info: `module spider <name>`.
- Load a module: `module load <name>`.
- Unload a module: `module unload <name>`.
- Unload all modules: `module purge`.

Note: Some modules need other modules to be loaded first. These requirements are specified in the `module spider` info.

Useful modules for this course: GCC, tbb, NVHPC, OpenMPI, ...

Job: Script SLURM (1)

Job scheduling done via SLURM.

The job's properties are defined in the script:

```
#SBATCH --job-name=parallel_class_job
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --time=00:10:00
#SBATCH --partition=shared-cpu
#SBATCH --mem=6000M
```

Job: Script SLURM (2)

In addition for this course:

```
# Using reservation made for this course
#SBATCH --account=latt_teach_parallelisme
#SBATCH --reservation=...
```

Other properties:

```
# For a GPU job, also use --partition=shared-gpu
#SBATCH --gpus=1

#SBATCH --output=slurm-%j.out
#SBATCH --error=slurm-%j.err
```

Job: Script SLURM (3)

Load the modules inside the script.

Possible to run the command via `srun`:

```
srun g++ -O3 main.cpp -o main
srun ./main
```

Sets up a context for the parallel execution.
Gives a better history of the execution (`sacct`).

Compiling inside the job: good practice for heavy compilations.

IMPORTANT
No code must be run on the login node.

Job: submit

Submit a job: `sbatch <script name>`

See your active/waiting jobs: `squeue -u $USER`

See the scheduling of a job:
`squeue -u $USER --start`

See the jobs done: `sacct`
With more details: `sac`

Cancel a job: `scancel <id>`